

POLS 3316: STATISTICS FOR POLITICAL SCIENTISTS

Fall 2026 | Class No. 13875

Department of Political Science, University of Houston

Syllabus Version 4: September 22, 2026

Instructor:	Songeun (Emily) Lee	Time:	TuTh 11:30 AM–1:00 PM
Email:	selee5@cougarnet.uh.edu	Place:	PGH 200
Office Hours:	Wednesdays, 11:00 AM–12:00 PM via Teams, or by appointment.		

Course Description

How do voters decide whom to support? Why do some policies work better than others? This course introduces the basic tools political scientists use to answer these questions. We will learn how to develop research questions and theories, think about cause and effect, measure political concepts, and use data to test arguments. Topics include research design, measurement, descriptive statistics, probability, hypothesis testing, and regression. We will also cover multiple regression, limited dependent variables, and time-series data. Class examples will use real political questions, along with simple formulas, tables, graphs, and calculations. This is not a programming course, and no statistical software is required. However, students should be prepared to work with numbers and data throughout the semester.

Objectives

By the end of the semester, you should be able to:

1. Explain the scientific approach to studying politics and develop clear research questions.
2. Identify variables, causal claims, and basic theoretical relationships.
3. Apply the four causal hurdles to evaluate causal claims.
4. Distinguish experimental from observational research designs and explain their trade-offs.
5. Evaluate the reliability and validity of measures of political concepts.
6. Identify variable types and compute and interpret descriptive statistics.
7. Use probability theory, the central limit theorem, bivariate hypothesis tests, and basic regression methods, including multiple regression and extensions for limited dependent variables and time-series data.

Prerequisites

MATH 1314, GOVT 2306, and GOVT 2305 or equivalent, or consent of instructor.

Required and Recommended Textbooks

Required.

- Kellstedt, Paul M. and Guy D. Whitten. 2018. *The Fundamentals of Political Science Research*. 3rd ed. Cambridge University Press.

Recommended (optional).

- Pollock III, Philip H. 2015. *The Essentials of Political Analysis*. 5th ed. CQ Press. A friendly supplementary text covering many of the same concepts at an introductory level. Useful if you want an alternative explanation of a topic from class.

Grading

Final grades are based on the weighted total of the components below.

Component	Weight
Attendance and Participation	10%
Homework assignments (4 assignments, 10% each)	40%
Midterm exam	25%
Final exam	25%
Total	100%

Attendance and Participation

Attendance will be taken on randomly selected class days, so please arrive before class begins. You may miss one class without penalty. After the one free absence, your attendance grade will be based on the percentage of remaining attendance checks you complete. Absences will be excused only for reasons covered by UH policy, and excused absences will not reduce your attendance grade. If you miss class, you are responsible for catching up on readings, lecture material, assignments, and announcements.

Homework Assignments

There are four homework assignments this semester. Each homework assignment is worth 10% of your final course grade. The questions will cover definitions, key concepts, examples, interpretations, and basic mathematical calculations from class. Some calculation questions may ask you to show your work, so be sure to include the steps you used to get your answer.

Homework must be submitted on Canvas by the posted deadline. Unless I announce otherwise, please submit your homework as a PDF or Word document. I will not accept homework submissions by email. Because internet or Canvas issues can happen, I strongly recommend uploading your assignment well before the 11:59 p.m. deadline. Handwritten work is allowed, but it must be clearly readable; if I cannot read your answer, it may receive no credit. You may use the textbook, your notes, and course materials while completing homework, but each student must submit their own work, and answers must be written in your own words.

Late Submission

Students may submit one homework assignment late during the semester without penalty. After this one-time exception has been used, late homework assignments will receive a 20-point deduction on the 100-point scale. I will waive the late penalty for additional late submissions only for reasons covered by UH policy, and documentation may be required.

Late homework assignments may still be submitted after the original deadline. To allow enough time to calculate final grades, all late homework assignments must be submitted by Thursday, December 3, 2026, at 11:59 PM.

Exams

The midterm and final exams will cover concepts and basic calculations from the assigned textbook chapters, class lectures, and homework. Exams are not cumulative, although later topics may build on earlier material.

Both exams are open-book and open-note. You may use the textbook, notes, class slides, homework, printed course materials, and a basic standalone calculator. Phones, laptops, tablets, smart watches, AI tools,

internet access, and communication with others are not permitted. Calculations must be done using a standalone calculator, not a phone, computer, or other internet-connected device. Unauthorized assistance or materials constitute an academic integrity violation.

The midterm will be held in person on Thursday, October 15, 2026, and the final exam on Thursday, December 3, 2026. The Tuesday class before each exam will be devoted to review. Make-up exams are permitted only for reasons covered by UH policy and may require documentation; travel plans and forgetting the exam date do not qualify.

Extra Credit

You may earn up to 5 extra credit percentage points, added to your final course grade. Extra credit cannot replace required homework, exams, or attendance.

To earn extra credit, choose one or two of the articles listed below. For each article, submit a one-page summary and a one-page response. Each completed article is worth up to 2.5 percentage points.

In your response, answer the questions below. The listed Kellstedt and Whitten chapters may be helpful as you think through these questions.

1. What is the article's main research question? *See Chapters 1–2.*
2. What are the main independent variable, dependent variable, and key control variables? *See Chapters 2 and 6.*
3. How are the main variables measured? *See Chapters 5–6.*
4. What research design does the article use, and what kind of causal claim can the authors make? *See Chapters 3–4.*

Submit your write-up as a PDF or Word document on Canvas.

Article options (posted on Canvas):

1. Anzia, Sarah F. and Christopher R. Berry. 2011. "The Jackie (and Jill) Robinson Effect: Why Do Congresswomen Outperform Congressmen?" *American Journal of Political Science*.
2. Bauer, Nichole M. 2020. "Shifting Standards: How Voters Evaluate the Qualifications of Female and Male Candidates." *Journal of Politics*.
3. Corbett, Christianne, Jan G. Voelkel, Marianne Cooper, and Robb Willer. 2022. "Pragmatic Bias Impedes Women's Access to Political Leadership." *Proceedings of the National Academy of Sciences*.
4. Jenke, Libby. 2024. "Affective Polarization and Misinformation Belief." *Political Behavior* 46: 825–884.

Extra credit is due by Thursday, December 3, 2026, at 11:59 PM. Late extra credit will not be accepted.

Important Dates

Date	Important Course Event
Sept. 15	Homework 1 due by 11:59 PM.
Oct. 13	Homework 2 due by 11:59 PM.
Oct. 15	Midterm Exam.
Nov. 10	Homework 3 due by 11:59 PM.
Dec. 1	Homework 4 due by 11:59 PM.
Dec. 3	Final Exam. Extra credit and any late homework submissions due by 11:59 PM.

Course Timeline

The course schedule is subject to change. Any changes will be announced through Canvas. Please read the assigned chapter before coming to class.

Part I. Foundations of Scientific Inquiry

Week 1
Aug. 25 & 27

Course Introduction and the Scientific Study of Politics
Aug. 25 Course introduction, syllabus walk-through, and course expectations.
Aug. 27 The scientific study of politics: facts, values, and causal explanations.
Reading. Kellstedt & Whitten, Chapter 1, Sections 1.1–1.6.

Week 2
Sept. 1 & 3

The Art of Theory Building
Sept. 1 Research questions, good theories, and thinking about causes.
Reading. Kellstedt & Whitten, Chapter 2, Sections 2.1–2.9.
Sept. 3 No class – APSA Annual Meeting.
Posted. Homework 1 (Chapters 1–3) posted on Canvas.

Part II. Causality and Research Design

Week 3
Sept. 8 & 10

Evaluating Causal Relationships
Sept. 8 Causality; the four hurdles along the route to establishing causal relationships.
Reading. Kellstedt & Whitten, Chapter 3, Sections 3.1–3.2.
Sept. 10 Why studying causality is important: examples from political science.
Reading. Kellstedt & Whitten, Chapter 3, Sections 3.3–3.4.

Week 4
Sept. 15 & 17

Research Design (Part 1): Experiments
Sept. 15 Causal comparison, experimental design, and random assignment.
Reading. Kellstedt & Whitten, Chapter 4, Sections 4.1–4.2.2.
Due. Homework 1 (Chapters 1–3) due today by 11:59 PM.
Sept. 17 Types of experiments and limits of experimental research.
Reading. Kellstedt & Whitten, Chapter 4, Sections 4.2.3–4.2.4.

Week 5
Sept. 22 & 24

Research Design (Part 2): Observational Studies and Design Trade-offs
Sept. 22 Cross-sectional and time-series research designs.
Reading. Kellstedt & Whitten, Chapter 4, Sections 4.3–4.3.3.
Sept. 24 Comparing research designs: internal and external validity, trade-offs, and applications in political science.
Reading. Kellstedt & Whitten, Chapter 4, Sections 4.3.4–4.5.
Reading. McDermott, Rose. 2002. “Experimental Methods in Political Science.” *Annual Review of Political Science* 5: 31–61.
Reading. Miller, Michael G. and Joseph L. Sutherland. 2023. “The Effect of Gender on Interruptions at Congressional Hearings.” *American Political Science Review* 117(1): 103–121.

Part III. Measurement

Week 6 Measuring Concepts of Interest

Sept. 29 & Oct.
1

Sept. 29 Getting to know your data; measurement challenges; reliability, and validity.
Reading. Kellstedt & Whitten, Chapter 5, Sections 5.1–5.3.

Oct. 1 Measuring democracy and political attitudes; consequences of measurement choices.
Reading. Kellstedt & Whitten, Chapter 5, Sections 5.4–5.7.

Reading. Druckman, James N. and Matthew S. Levendusky. 2019. “What Do We Measure When We Measure Affective Polarization?” *Public Opinion Quarterly* 83(1): 114–122.

Posted. Homework 2 (Chapters 4–6) posted on Canvas.

Part IV. Descriptive Statistics

Week 7 Getting to Know Your Data

Oct. 6 & 8

Oct. 6 Variable types (categorical, ordinal, continuous) and describing categorical variables.
Reading. Kellstedt & Whitten, Chapter 6, Sections 6.1–6.3.

Oct. 8 Describing continuous variables: rank statistics, and limitations of descriptive statistics.
Reading. Kellstedt & Whitten, Chapter 6, Sections 6.4–6.6.

Midterm Exam Week

Week 8 Midterm Exam

Oct. 13 & 15

Oct. 13 Midterm review session.

Due. Homework 2 (Chapters 4–6) due today by 11:59 PM.

Oct. 15 Midterm Exam.

Part V. Probability and Statistical Inference

Week 9 Probability and Statistical Inference

Oct. 20 & 22

Oct. 20 Populations, samples, probability, and the logic of statistical inference.
Reading. Kellstedt & Whitten, Chapter 7, Sections 7.1–7.2.

Oct. 22 The Central Limit Theorem, the normal distribution, and sampling in practice.
Reading. Kellstedt & Whitten, Chapter 7, Sections 7.3–7.5.

Part VI. Bivariate Hypothesis Testing

Week 10 Bivariate Hypothesis Testing

Oct. 27 & 29

Oct. 27 Bivariate hypothesis tests, choosing the right test, and the logic of p -values.
Reading. Kellstedt & Whitten, Chapter 8, Sections 8.1–8.3.

Oct. 29 Difference of means, correlation, and statistical significance.
Reading. Kellstedt & Whitten, Chapter 8, Sections 8.4–8.5.

Posted. Homework 3 (Chapters 7–9) posted on Canvas.

Part VII. Two-Variable Regression

Week 11

Nov. 3 & 5

Two-Variable Regression Models**Nov. 3** Two-variable regression, fitting a line, and estimating the OLS regression line.*Reading.* Kellstedt & Whitten, Chapter 9, Sections 9.1–9.3.**Nov. 5** Model fit, uncertainty, confidence intervals, hypothesis tests, and OLS assumptions.*Reading.* Kellstedt & Whitten, Chapter 9, Sections 9.4–9.5.

Part VIII. Multiple Regression

Week 12

Nov. 10 & 12

Multiple Regression: The Basics**Nov. 10** Introduction to multiple regression and interpreting coefficients.*Reading.* Kellstedt & Whitten, Chapter 10, Sections 10.1–10.4.*Due.* Homework 3 (Chapters 7–9) due today by 11:59 PM.**Nov. 12** Statistical significance, omitted-variable bias, and presenting results.*Reading.* Kellstedt & Whitten, Chapter 10, Sections 10.5–10.10.

Part IX. Multiple Regression Model Specification

Week 13

Nov. 17 & 19

Multiple Regression Model Specification**Nov. 17** Dummy independent variables and interactive hypotheses in OLS regression.*Reading.* Kellstedt & Whitten, Chapter 11, Sections 11.1–11.3.**Nov. 19** Outliers, influential cases, multicollinearity, and model specification.*Reading.* Kellstedt & Whitten, Chapter 11, Sections 11.4–11.6.*Posted.* Homework 4 (Chapters 10–12) posted on Canvas.

Part X. Extensions of Regression

Week 14

Nov. 24 & 26

Limited Dependent Variables and Time-Series Data**Nov. 24** Dummy dependent variables, limited dependent variable models, and an introduction to time-series data.*Reading.* Kellstedt & Whitten, Chapter 12, Sections 12.1–12.5.**Nov. 26** *Thanksgiving holiday — no class.*

Final Exam Week

Week 15

Dec. 1 & 3

Review and Final Exam**Dec. 1** Final exam review session.*Due.* Homework 4 (Chapters 10–12) due today by 11:59 PM.**Dec. 3 Final Exam.** Extra credit and any late homework submissions due by 11:59 PM.

Grading Scale

Final grades will be rounded to the nearest whole number before letter grades are assigned. A decimal of 0.5 or higher will be rounded up to the next whole number, while a decimal below 0.5 will be rounded down. For example, 89.4 will be rounded to 89, while 89.5 will be rounded to 90. There will be no exceptions to the grading scale or rounding policy.

If you have a question about a grade, please contact me within one week after the grade is posted. Regrade requests must clearly explain what you believe was graded incorrectly. I will review the relevant assignment or exam item. Requests made at the end of the semester to raise a final grade will not be considered unless there is a clear grading or calculation error.

After rounding, final letter grades will be assigned according to the scale below.

Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
Percentage	93–100	90–92	87–89	83–86	80–82	77–79	73–76	70–72	67–69	63–66	60–62	Below 60

Course Communication and Policies

Canvas

Course materials, announcements, homework assignments, grades, and submissions are handled through Canvas. Check Canvas regularly. Not seeing an announcement is not an excuse for missing a deadline.

UH Email

Please check and use your CougarNet email for course-related communication. If you email me about this course, include **POLS 3316** in the subject line. I will use CougarNet email for course-related matters, including grades and progress questions, because of FERPA rules. To access your CougarNet email, log in to Microsoft 365 with your CougarNet credentials. For help connecting CougarNet email to your phone or device, visit University Information Technology (UIT).

Communication

Please keep all course communication respectful and professional, including class discussions, Canvas messages, office hours, and email. You are welcome to ask questions, share opinions, and disagree with others, as long as you do so respectfully. I generally respond during weekday working hours.

Use of Artificial Intelligence (AI)

You may use AI tools as a tutor to help you learn. For example, you may ask AI to explain an unfamiliar term, give a simple example, or help you understand why a formula works. For instance, if we learn about p-values, you may ask AI, “Can you explain p-values in simple language?” I encourage this kind of use.

However, AI may not replace your own work. Homework answers, written interpretations, calculation explanations, exam answers, and summaries must be written in your own words and reflect your own understanding. Copying and pasting AI-generated answers is not allowed. If you use AI or any outside source to help prepare an assignment, you must clearly acknowledge or cite it when appropriate.

You are responsible for any mistakes that come from AI use. Assignments may be reviewed using Canvas-related AI detection tools, and work that appears to rely on AI-generated text rather than your own understanding may be reported as a possible academic integrity violation. If you are unsure whether a specific use of AI is allowed, please ask me before submitting the assignment.

Syllabus Changes

The weekly schedule may change if needed. Any changes to readings, assignments, or deadlines will be announced through Canvas.

Campus Closure and Emergency Changes

If the university closes campus or moves classes online, I will announce any course changes through Canvas.

Recording of Class

Class sessions will not be recorded or posted on Canvas. Students may not record, livestream, screenshot, or share any part of class without my advance written consent.

If you have or think you may have a disability-related need to record class activities, please contact the Justin Dart, Jr. Student Accessibility Center. Any approved recordings are for personal academic use only and may not be shared with any other person or posted on any platform.

Failure to follow this policy may result in a disciplinary referral to the Dean of Students Office.

University Policies and Resources

Add/Drop and Withdrawal Deadlines

You are responsible for knowing and following all University deadlines for adding, dropping, withdrawing from, or changing the grade status of this course. These deadlines are set by the University, not by the instructor. Please check the official calendar for the current semester: [UH Fall 2026 Academic Calendar](#).

Academic Honesty

All students must follow the UH Academic Honesty Policy. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, submitting work that is not your own, and using unauthorized resources during exams or assignments. This policy also applies to the AI policy described above. Violations may result in course penalties and referral to the university academic honesty process.

Excused Absences

Absences may be excused under the UH Excused Absence Policy. Covered reasons include illness, death of a close family member, required legal or government proceedings, university-sponsored activities, religious holy days, military service, pregnancy-related conditions, disability-related needs, or other reasons covered by university policy. If you have an excused absence, you will have a chance to make up work that counts toward the course grade or receive a reasonable alternative. Contact me as soon as possible and provide documentation when required.

Reasonable Academic Adjustments / Auxiliary Aids

UH is committed to providing an accessible learning environment. If you experience barriers to learning, assessment, or participation because of a disability, please contact the Justin Dart, Jr. Student Accessibility Center as soon as possible. Students seeking academic accommodations should register with the Dart Center early so approved accommodations can be implemented in a timely manner. Visit <https://www.uh.edu/accessibility>, call 713-743-5400, or email JDCenter@central.uh.edu.

Mental Health and Wellness

The University of Houston offers several resources to support student mental health and wellness, including CAPS, CoogsCARE, the UH Go App, the Student Health Center, the A.D. Bruce Religion Center, CSAC, the Women and Gender Resource Center, and the Cougar Parent and Family Engagement Office.

UH Counseling and Psychological Services (CAPS) offers counseling, workshops, group therapy, and informal Let's Talk consultations. CAPS also provides 24/7 crisis support by phone. Visit <https://www.uh.edu/caps> or call 713-743-5454. If you or someone you know is in crisis, call CAPS crisis support at 713-743-5454 or call/text the National Suicide and Crisis Lifeline at 988.

The Student Health Center offers a Psychiatry Clinic for enrolled UH students. To schedule an appointment, call 713-743-5149 during clinic hours, Monday through Friday, 8:00 a.m. to 4:30 p.m.

The A.D. Bruce Religion Center, CSAC, the Women and Gender Resource Center, and the Cougar Parent and Family Engagement Office also provide support for student well-being, basic needs, advocacy, family responsibilities, and campus connection.

Title IX / Sexual Misconduct

Under the UHS Sexual Misconduct Policy, your instructor is a responsible employee for Title IX reporting. This means I must report incidents of sexual misconduct to the Title IX office, known at UH as the Equal Opportunity Services office. Sexual misconduct includes sexual harassment, non-consensual sexual contact, sexual assault, sexual exploitation, sexual intimidation, intimate partner violence, and stalking.

Confidential resources are available on campus for students who want to speak privately. Reporting options and resources are available here: <https://uh.edu/equal-opportunity/title-ix-sexual-misconduct/resources/>.